

NTT at Real Scale: $n = 256$

The $n=8$ example showed the full process, every single step, because it was small enough to fit on a page. At real Kyber size ($n=256$, $q=3329$, $\text{zeta}=17$), the SAME exact process happens - just far more of it. This document shows the scale, the real magic numbers, and a real excerpt of the very first few steps - not the whole thing, since a complete trace would be roughly 1,024 individual butterflies long.

The scale, compared to the small example

$n=8$ example: 2 rounds, 6 butterflies total

$n=256$ real: 8 rounds, 1,024 butterflies total (8 rounds $\times$ 128 per round)

Same idea, same mechanism (one multiply, then add/subtract) - just repeated far more times, since there are 256 numbers to process instead of 8.

The real magic number chain (starting from $\text{zeta}=17$, mod 3329)

Same rule as before: start with 17, keep multiplying by 17 and reducing mod 3329.

link 1: 1729

link 2: 2580

link 3: 3289

link 4: 2642

link 5: 630

link 6: 1897

link 7: 848

link 8: 1062

... this chain continues for 127 links total ($n/2 - 1 = 127$), same idea throughout.

A real excerpt: the first 3 butterflies of Round 1

Example polynomial (only a few non-zero coefficients shown, rest are 0 for readability):

position 0 = 5, position 1 = 12, position 2 = 300

position 128 = 40, position 129 = 900, position 130 = 3

Round 1 pairs position j with position $j+128$ (same shape as the $n=8$ example pairing j with $j+4$ - just a bigger jump, since $n/2=128$ here instead of $n/2=4$).

Magic number for Round 1: 1729

Pair (position 0, position 128): values 5, 40

$$t = 1729 \times 40 \bmod 3329 = 2580$$

$$5 + 2580 \bmod 3329 = 2585 \rightarrow \text{position 0}$$

$$5 - 2580 \bmod 3329 = 754 \rightarrow \text{position 128}$$

Pair (position 1, position 129): values 12, 900

$$t = 1729 \times 900 \bmod 3329 = 1457$$

$$12 + 1457 \bmod 3329 = 1469 \rightarrow \text{position 1}$$

$$12 - 1457 \bmod 3329 = 1884 \rightarrow \text{position 129}$$

Pair (position 2, position 130): values 300, 3

$$t = 1729 \times 3 \bmod 3329 = 1858$$

$$300 + 1858 \bmod 3329 = 2158 \rightarrow \text{position 2}$$

$$300 - 1858 \bmod 3329 = 1771 \rightarrow \text{position 130}$$

This exact process (one multiply, one add, one subtract) repeats for the remaining 125 pairs of Round 1, then again for all of Rounds 2 through 8 - same mechanism, just a lot more of it. Not shown in full here since it would just be the same arithmetic pattern copied roughly 1,020 more times.

Confirming it all still works at real scale

Running the complete real $n=256$ pipeline (NTT.NTT_Multiplication.py) - transforming two full 256-coefficient polynomials, multiplying them the fast way, transforming back, and comparing to the slow schoolbook answer:

schoolbook and NTT results match: True

first 5 coefficients, schoolbook: [2720, 381, 397, 2480, 2871]

first 5 coefficients, NTT method: [2720, 381, 397, 2480, 2871]

Summary

Real Kyber's NTT is exactly the $n=8$ process you already worked through by hand - the same butterfly mechanism, the same idea of a magic-number chain built from one starting value - just scaled up to 256 numbers and 8 rounds instead of 8 numbers and 2 rounds. Nothing new happens conceptually between the small example and the real thing; there is simply far more of the same arithmetic repeated.